

Wireshark

- Has a **GUI (Graphical User Interface)**
- **User-friendly** and easy to understand
- Runs on **Windows, Linux, and macOS**
- Packets can be **searched and filtered** by port number, source/destination IP, protocol, and time
- Because it is GUI-based, users can **click and analyze packets visually**, making it suitable for beginners

Similarities

- Both are **packet sniffers**
- Both are used to **capture and analyze network traffic**
- Both are **open-source tools**
- Both help in **network troubleshooting and security analysis**

tcpdump

- **Command-line (CLI) based**
- **Harder to understand** for beginners
- Runs mainly on **Linux/Unix systems**
- Used to **capture live network traffic**
- Captured packets can be **saved to a file** and later analyzed (for example, in Wireshark)
- **Commands must be memorized**, which makes it less user-friendly
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